THOMAS HOWARD

CTO & HEAD OF APPLIED AI | LLM SYSTEMS | AGENTIC AI | TECHNICAL LEADERSHIP

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PROFESSIONAL SUMMARY

Applied AI engineering leader specializing in **LLM applications, conversational agents, and governed autonomous systems**. I architect agent orchestration, tool execution, context management, structured outputs, and evaluation, then build the verification and authorization controls around model-driven workflows. At Snap eHealth, I lead healthcare AI product engineering and autonomous software delivery, including **Sally** and **Symphony-based agent orchestration**. I bring more than a decade of engineering leadership and hands-on Python, Go, and TypeScript development across distributed software systems.

// EXPERIENCE

Chief Technology Officer & Head of Applied AI | Snap eHealth

Jun 2026 – Present

Own Applied AI strategy, product architecture, and engineering delivery for a healthcare SaaS company, combining technical leadership with hands-on development.

- Lead development of **Sally, an LLM-powered healthcare voice agent**, combining real-time conversation, tool execution, background analysis, and structured visit-brief extraction.
- Define boundaries between model-generated behavior and **deterministic verification, consent, authorization, and tool access** in patient-facing workflows.
- Develop **evaluation harnesses and scenario-based tests** for conversational behavior, structured outputs, safety rules, and failure handling as part of Sally's production hardening.
- Architect multi-agent workflows using **Codex and Symphony**; coordinate **10 specialized engineering agents** across implementation, testing, review, debugging, and remediation.
- Design **isolated execution, tool-permission boundaries, automated verification, auditability, and model-cost controls**, with human authorization over merges and production changes.
- Led delivery of **four production applications in four months** across enterprise healthcare, workflow automation, and patient engagement.
- Lead senior engineers through architecture review and AI-assisted delivery; remain hands-on in **Python, TypeScript, APIs, and healthcare integrations**, including debugging and code review.
- Set the **Applied AI roadmap and technical priorities** with the CEO and Product, translating customer needs into agent capabilities, architecture decisions, and release criteria.

Head of AI Platform & Infrastructure | Snap eHealth

Oct 2022 – Jun 2026

Led AI-assisted engineering, model evaluation, and healthcare application architecture, expanding technical ownership into company-wide Applied AI leadership.

- Led AI reliability initiatives using **Amazon Bedrock**, including **LLM evaluation workflows, goldset reference datasets, and guardrails**.
- Built an **AI-driven test-intelligence system** that decomposed product stories, generated test requirements and API tests, analyzed defects, and created targeted retests.
- Designed an **AI engineering control plane** with specialized agents, structured evidence, deterministic triage, bounded remediation, audit history, and human escalation.
- Built **continuous QA and SRE agents** to detect defects, security risks, platform regressions, and incidents, turning findings into actionable engineering work.

- Established **protected verification gates and human review** for agentic workflows, keeping automated execution bounded and remediation subject to engineering controls.
- Architected a physician referral prediction system using Python and scikit-learn, incorporating provider quality, patient eligibility, specialty alignment, appointment availability, and historical outcomes to support data-driven referral decisions.
- Designed an explainable physician-ranking engine combining clinical quality, provider experience, outcomes, network compatibility, geographic proximity, and evidence confidence, with deterministic scoring, uncertainty handling, and abstention safeguards.
- Developing supervised machine learning workflows to predict 30-day referral completion, incorporating feature engineering, classification models, data-leakage prevention, and evaluation using precision, recall, ROC-AUC, and probability calibration.
- Integrated predictive analytics and decision-support architecture into healthcare referral workflows, emphasizing interpretable recommendations, data quality, and controlled clinical decision boundaries.

Founder & Software Engineer | BusyNow

Oct 2025 – Present

Founded and operate **BusyNow**, a consumer SaaS product; own product strategy, architecture, implementation, and production operations.

- Built a **React/TypeScript frontend and Node.js/Express backend**, integrating REST APIs, PostgreSQL, and Redis into a live application.
- Use **Codex and supervised agent workflows** for architecture, implementation, test generation, debugging, and review, with CI verification and human-controlled releases.
- Designed transactional workflows with **idempotency, retries, and data-correction paths** to support reliable application behavior under partial failure.
- Implemented **dependency isolation, caching, rate limits, and provider budgets** to contain external-service failures and control operating costs.
- Engineered the **AWS production runtime** with ECS, Terraform, VPC, IAM, CloudFront, S3, and containerized application services.
- Implemented **ECS deployment workflows** with health checks and rollback for repeatable releases and operational recovery.

Director of Engineering – DevOps | FBS

Jan 2015 – Apr 2022

Led a **12-person engineering organization** supporting enterprise software delivery, cloud architecture, and production infrastructure across multiple product lines.

- Built and led the organization through an engineering manager, owning **hiring, technical direction, mentoring, performance management, and leadership development**.
- Reduced regression cycle time by **approximately 90%** through an automation-first transformation, earlier validation, and repeatable release controls.
- Directed **CI/CD architecture, developer tooling, deployment automation, and engineering priorities** in partnership with product and engineering leadership.
- Architected **Jenkins delivery pipelines** for build automation, integration, artifact promotion, environment consistency, and deployment.
- Designed and operated **AWS environments** using VPC, EC2, RDS, IAM, Docker, and ECS to support scalable enterprise applications.
- Established **production observability and operational practices** with Grafana, CloudWatch, ELK, and New Relic, improving visibility into application health.

EARLIER EXPERIENCE

Software Engineer, Automation Infrastructure | **Google (Contract)** Jun 2012 – Dec 2014

Software Engineer | **Community Health Plan of Washington** Aug 2011 – Jun 2012

Software Engineer | **Xperience Communications** Aug 2010 – Aug 2011

// SELECTED PROJECTS

Sally — Conversational Healthcare Agent

LLM-powered voice workflows combining browser and phone interaction, interruption handling, background analysis, and structured visit briefs. Engineering includes **scenario-based evaluation**, consent and verification boundaries, and deterministic controls over authorized actions.

Symphony — Governed Autonomous Engineering

Agent orchestration for engineering tasks with **isolated execution**, repository-scoped context, signed leases, tool permissions, cancellation, cost accounting, and audit trails. Staged validation and verification gates retain human authority over review, merge, and deployment.

AI Engineering Control Plane & Evaluation Harness

Reusable **specialized-agent workflows** for quality, security, reliability, and release readiness. Structured findings feed deterministic triage, bounded remediation, protected verification, and human escalation rather than unreviewed production changes.

Referral Intelligence — Predictive ML & Physician Recommendation

Healthcare decision-support platform combining **explainable physician ranking, clinical outcomes, provider quality, and referral intelligence**. Leading the development of predictive ML capabilities to train models on historical referral outcomes, estimate referral success probabilities, and improve physician recommendations through measurable model evaluation and validation.

// SKILLS

AI / Agentic: LLM application architecture, conversational AI, agent orchestration, tool use, context engineering, structured outputs, Codex, Symphony

Evaluation / Safety: Agent evaluations, regression harnesses, deterministic verification, tool authorization, execution isolation, provenance, bounded remediation, human approval

Engineering: Python, Go, TypeScript, React, Node.js/Express, REST APIs, PostgreSQL, Redis, Playwright, distributed systems

Cloud / Production: AWS, Kubernetes, EKS, Terraform, Helm, Argo CD, Argo Rollouts, CI/CD, observability, reliability engineering

Leadership: Applied AI strategy, product architecture, technical roadmaps, engineering leadership, mentoring, customer discovery, executive communication